

HPC ENVIRONMENT & SIMULATION GUIDE

Custom Operational Manual for High-Statistics Radiation Transport Modeling

This streamlined operational handbook consolidates technical protocols, shell workflows, and automated parameters established for your remote SSH high-performance computing (HPC) environment. It isolates runtime configurations specifically required for optimizing large-scale parallel Monte Carlo particle simulations.

1. Environment Automation & G4Data Paths

To eliminate manual resource declarations during remote sessions, the system initialization path variables have been anchored to target your active dependencies automatically across all working directories.

Persistent Environment Exports

The following declarations are configured within your remote shell resource profile (`~/.bashrc` or `~/.zshrc`) to universally handle Geant4 data mapping and executable lookups:

```
# TOPAS and Geant4 Data Configuration
export TOPAS_G4_DATA_DIR="/net/home/samt/topas/G4Data"

# Append Binary Directory to Active Executable PATH
export PATH="/net/home/samt/topas/bin:$PATH"
```

Runtime Sync: To force immediate propagation of environment path alignments within an existing SSH session window, execute: `source ~/.bashrc`

2. High-Statistics Multi-Threading Calculations

Scaling up to high-statistics runs such as $N = 10^9$ (1 Billion Histories) drives the relative statistical uncertainty down via parallel variance tracking. Optimizing processing time requires matching parameter thread weights to hardware boundaries.

Logical Processing Array Ceiling

Hardware diagnostic interrogation (`nproc`) reports a cluster ceiling of `64 logical cores` . To manage high workloads efficiently:

- **Recommended Production Threshold:** Allocate `60 threads` . This utilizes 93.75% of computational cores for parallel transport calculation while keeping 4 threads uninhibited to mitigate I/O latency, execute background tasks, and stabilize remote terminals.
- **Parameter Declarations:** Configure these entries within your runtime script macro (e.g., `UITF_ND3.txt`):

```
i:Ts/NumberOfThreads    = 60
i:Ts/NumberOfHistories  = 1000000000
```

Suppression of Structural Overhead

To maximize track iteration throughput and prevent memory saturation at scale, suppress visualization engines and throttle standard terminal stream updates:

```
# Complete Suppression of Graphics Engines
b:Ts/UseGraphics = "False"

# Throttling Standard Terminal Stream I/O
i:Ts/ShowHistoryCountAtInterval = 1000000 # Log intervals limited to 1M events
i:Ts/TrackingVerbosity = 0                # Complete logging suppression
```

3. Detached Process Execution and Disowning

To guarantee that extensive Monte Carlo tasks survive unexpected SSH session decay, terminal closures, or local machine disconnects, simulations must be completely decoupled from the parent login process tree.

The Independent Execution Pipeline

Initialize the simulation inside the system background array, pipe stdout and stderr tracking to a localized log structure, and strip the terminal hangup signal dependency:

```
# Step 1: Start simulation asynchronously and pipe telemetry output
topas UITF_ND3.txt > simulation.log 2>&1 &

# Step 2: Detach running job from current terminal session
disown
```

Safe Terminal Disconnection

Once the process is disowned, it runs as an independent daemon thread on the server. Safely terminate your active shell workspace via:

```
exit
```

4. Remote Interrogation & Process Monitoring

To verify structural health or modify parallel worker streams upon establishing a new SSH session connection, execute these primary terminal utilities:

Stream Verification

Read log telemetry pipelines to monitor the active iteration cycles of individual Geant4 workers (e.g., `G4WT#` tracking ticks):

```
# Output final 20 static telemetry updates
tail -n 20 simulation.log

# Stream processing log adjustments dynamically in real-time
tail -f simulation.log
```

Operation Note: Entering `Ctrl + C` breaks out of a live streaming `tail` view. This action safely severs the console view application only; it has no impact on the underlying simulation daemon.

Resource Allocations and Signal Controls

Analyze real-time core processing or stop run procedures if modifications are required:

```
# Launch interactive real-time core performance interface
htop

# Terminate all currently running instances of the topas calculation binary
pkill topas
```